

OpenTelemetry Industrial

Community Overview & Technical Vision

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github.com/otel-industrial



Agenda

01

OpenTelemetry Industrial

Community overview, mission & projects

02

Community Resources

GitHub, Slack, projects

03

Roadmap

Vision and future direction

04

Integration Models

How industrial data reaches OpenTelemetry

05

OpenTelemetry Collector

Architecture and data flow

06

Semantic Conventions

Standardizing industrial telemetry

What is OpenTelemetry Industrial community?

Community Status

Community-led initiative

We operate as a user group. The concept of 'user groups' is not yet formalized in OpenTelemetry (this is actively being discussed).

Monthly community calls, open to all.

[github.com/otel-industrial/
otel-industrial-community](https://github.com/otel-industrial/otel-industrial-community)

Founded May 2026

60+ members

CNCF Slack: **#otel-industrial**

A small but active core contributing receivers, docs, and community resources.

github.com/otel-industrial

Our Mission

Industrial Observability

Build an open, vendor-neutral ecosystem for industrial observability through OpenTelemetry, bringing together end users, vendors, and the open source community.

OpenTelemetry Industrial Resources

May 2026

Founded

60+

CNCF Slack members

#otel-industrial

CNCF Slack channel

GitHub

otel-industrial-resources

Documentation, reference materials, project catalog, and learning resources.

otel-industrial-community

Community meetings, roadmap, presentations, working groups, and collaboration.

Get Involved

CNCF Slack: #otel-industrial

Most active discussion happens here

Monthly community calls

Open to all — check GitHub for schedule

Contribute code, documentation, testing, and ideas

All skill levels welcome

OpenTelemetry Industrial Projects

PROJECT	DESCRIPTION	STATUS	AUTHOR
Modbus Receiver	OTel Collector receiver for Modbus devices and industrial equipment.	Experimental	lukaszciukaj
OPC UA Receiver	OTel Collector receiver for OPC UA servers and industrial systems.	Experimental	bruegth
MQTT Sparkplug Receiver	MQTT Sparkplug protocol support for OpenTelemetry Collector.	Active Dev	jmacd
mqtt2otel (bridge app)	MQTT-to-OTel bridge that transforms MQTT messages into OTel metrics and logs.	Experimental	OSgAgA
EtherNet/IP Receiver	OTel Collector receiver for EtherNet/IP	Work in Progress	Community

Where We Are Headed

Receivers for Popular Protocols

Stable OpenTelemetry Collector receivers for OPC UA, MQTT, Sparkplug, Modbus, EtherNet/IP, Profinet and other popular industrial protocols.

OTel Industrial Collector Distribution

A purpose-built OpenTelemetry Collector with industrial receivers pre-packaged. Easy to deploy at the factory edge.

Become an official part of OpenTelemetry

Formalize as an official part of OpenTelemetry project.

OTel Documentation for Industrial

A dedicated section on opentelemetry.io covering industrial use cases, onboarding guides, and protocol recipes.

Three Ways to Bring Industrial Data to OTel

Model 1

OTel SDK + OTLP

Industrial app or device uses the OTel SDK directly.

Sends telemetry via OTLP to a standard OTel Collector.

Best for modern, programmable edge devices with runtime SDK support.

Requires SDK integration in the application or device. Best suited for modern, programmable systems.

Model 2

Industrial Protocol Receivers

Dedicated receivers on the OTel Collector translate industrial protocols into OTel signals.

No SDK changes needed in devices. The Collector performs the translation.

Ideal for legacy or proprietary systems where SDK integration is not possible.

Model 3

Bridge Application

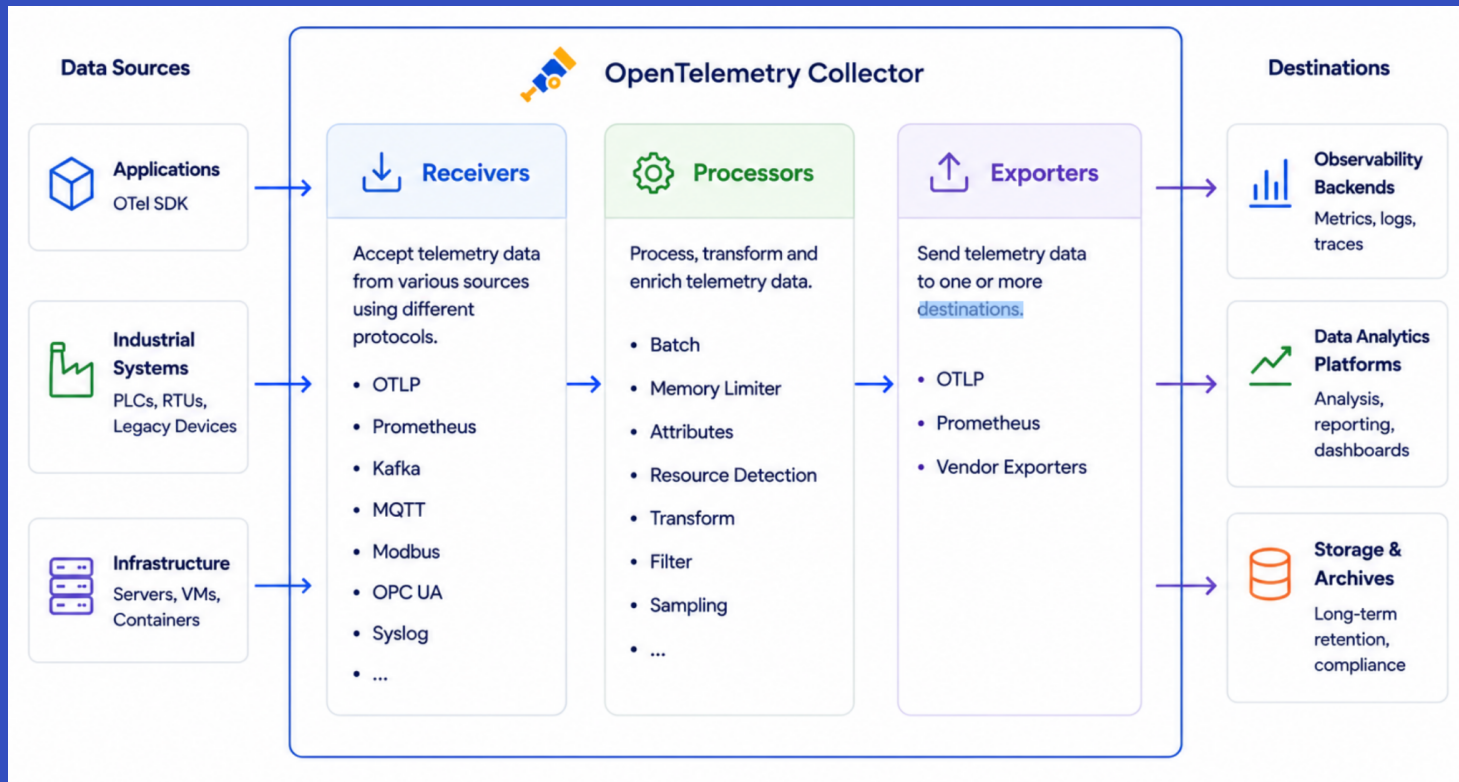
A separate bridge application reads industrial protocol data and converts it to OTLP.

Forwards to any standard OTel Collector.

Works with any backend, any protocol if library for industrial protocol is available.

Ideal when a native Collector receiver is unavailable or custom processing is required.

OpenTelemetry Collector



Industrial Semantic Conventions

What are Semantic Conventions?

A shared vocabulary for telemetry that defines what data is called, what it means, and how it is represented. Semantic conventions ensure telemetry is consistent and understandable across applications, systems, and observability tools.

Examples

Telemetry Concept	Semantic Convention
Service name	<code>service.name</code>
Service version	<code>service.version</code>
Host name	<code>host.name</code>
Cloud provider	<code>cloud.provider</code>
HTTP method	<code>http.request.method</code>

<https://opentelemetry.io/docs/specs/semconv/general/>

Why It Matters for OpenTelemetry Industrial

Enable vendor-neutral industrial telemetry

A common data model allows telemetry from different industrial protocols and vendors to be represented consistently.

Build reusable OpenTelemetry integrations

Receivers, processors, exporters, dashboards, and tooling can work across industrial technologies without protocol-specific customizations.

Build once, use everywhere

Defining conventions early helps avoid incompatible implementations as OpenTelemetry adoption grows across industrial environments.

Where We Need Your Help

1

Industrial Domain Expertise

We need contributors and reviewers with hands-on experience in industrial automation, OT environments, and industrial protocols to ensure high-quality implementations.

2

Vendor Resources & Testing Environments

We need access to industrial hardware, software, simulators, and vendor expertise to develop, validate, and test receivers and integrations in realistic environments.

3

End-user Adoption & Test Sites

We need organizations willing to provide local test environments for validating receivers and deployment patterns, and to share real-world feedback and use cases.

Industrial Observability 2026



Industrial Observability 2026

The first edition provides an overview of:

- Industry trends and challenges
- OpenTelemetry in industrial environments
- Reference architectures
- Community opportunities
- Future directions

Industrial Observability 2027

A community-driven whitepaper, likely maintained under the OpenTelemetry Industrial project.

Thank You!

Questions & Discussion

